UNITED STATES DISTRICT COURT
EASTERN DISTRICT OF LOUISIANA

KATHLEEN MOLLER, **Plaintiff,**	*	
v.	*	**CIVIL ACTION NO. 2:24-CV-228-WBV-DPC**
MARTIAN SALES, INC.; JOPEN, LLC; JOHNSON FOODS, LLC; LP IND., LLC; CAG HOLDINGS, LLC; RMH HOLDINGS, LLC; ABC INSURANCE COMPANY and John Does 1-4, **Defendants.**	*	**THE HONORABLE WENDY B. VITTER** **MAGISTRATE JUDGE DONNA PHILLIPS CURRAULT**

DEFENDANTS' MEMORANDUM IN SUPPORT OF MOTION TO EXCLUDE PLAINTIFF'S EXPERT, RONALD H. WHARTON MD, UNDER FED. RULE OF EVID. 702

Defendant JOpen, LLC ("Defendant" or "JOpen") hereby files this Motion to Exclude Plaintiff's Expert Witness, Ronald H. Wharton, MD, under Federal Rule of Evidence 702 and *Daubert v. Merrell Dow Pharmaceuticals, Inc.*, 509 U.S. 579 (1993) and the incorporated memorandum of law.

I. INTRODUCTION

Dr. Wharton's opinions—that the decedent, Harmony Moller, died from a cardiac event induced by kratom—are inadmissible for three reasons.

Plaintiff intends to present Dr. Ronald Wharton—a cardiologist with no background in toxicology, pharmacology, kratom research, or forensic pathology—to tell a jury that kratom caused Harmony Moller's death. But under controlling Fifth Circuit precedent, an expert must be qualified in the subject matter of his opinion, must establish general causation before offering specific causation, and must apply a reliable methodology at every step. Dr. Wharton satisfies none of those requirements.

First, he is not qualified to opine on kratom's toxic effects. He had never studied kratom, treated kratom exposure, researched kratom pharmacology, or encountered it clinically before this litigation. Rule 702 does not permit a cardiologist to step into the role of toxicologist simply because the case involves a heart.

Second, his causation theory collapses at the threshold. The Fifth Circuit requires proof of a harmful level of exposure before general causation may be presented to a jury. Dr. Wharton admits he does not know what a safe dose of mitragynine is, what a lethal dose is, or what exposure level is capable of causing fatal arrhythmia. Without that foundation, specific causation cannot proceed.

Third, his "process of elimination" is not a reliable differential diagnosis. He failed to investigate critical medical history, failed to account for numerous other substances, failed to review key materials, and relied on assumptions rather than verified facts. The Fifth Circuit prohibits that kind of post hoc reasoning.

Daubert requires intellectual rigor—not speculation, not litigation-generated hypotheses, and not conclusions reached because the expert "can't think of anything else." Because Dr. Wharton's testimony fails Rule 702 at each step—qualification, general causation, and specific causation—it must be excluded in its entirety.

II. FACTUAL BACKGROUND

On February 6, 2023, 36-year-old Harmony Moller was found dead in her home. Ex. 1 (November 18, 2025 Expert Report of Dr. Ron Wharton) ("Initial Report"), at 1. Ms. Moller's cause of death was listed as an "abnormally high left coronary ostia takeoff exacerbated by mitragynine use." *Id*. A toxicology report confirmed the presence of mitragynine at a level of 150 ng/mL in her femoral blood. *Id*.

This is a products liability case in which Plaintiff alleges that O.P.M.S. Silver kratom caused Ms. Moller's death. *See* ECF No. 100, at ¶¶ 30, 33. Specifically, Plaintiff claims that Defendants designed an inherently and unreasonably dangerous product, that Defendants failed to adequately warn of the dangers associated with O.P.M.S. Silver kratom, and that Defendants failed to provide adequate instructions on how to avoid such risks. *See* ECF No. 100, at ¶ 61.

Plaintiff retained Dr. Ronald Wharton, a cardiologist whose practice focuses on hypertrophic cardiomyopathy (HCM) (a condition that Dr. Wharton unequivocally states Harmony Moller did not have), to opine on the cause of death. Ex. 2, Excerpts of Deposition of Dr. Ronald Wharton ("Wharton Dep."), 47:23–49:24, Feb. 23, 2026.

Dr. Wharton admits he has no expertise in kratom or its toxicology or pharmacology. *Id.* at 27:4–28:15, 70:3-17; 100:3-7. He was not even "familiar with kratom prior to receiving this case." *Id.* at 43:1–47:13. He has never encountered kratom in his clinical practice and thus has not treated any patient with kratom exposure. Nor has he ever done any clinical research on kratom. *Id*. at 46:25–47:23. Dr. Wharton's "opinions on kratom in this case are based solely on the articles and references that [he] cited in his [his] report." *Id.* at 59:7-11. Apart from the articles cited in his report, Dr. Wharton testified he uses information he acquired from the internet to research kratom, and that this is a sufficient basis for him to offer expert testimony at trial on subjects such as the "profit margin" on kratom nationally. *Id.* at 62:5–65:25.

Dr. Wharton's has incomplete and incorrect knowledge on the O.P.M.S. Silver kratom product issue. He stated he "do[es] not know the specific product that she took" and "believe[s] it was a liquid" or an "extract." *Id.* at 66:1–67:17. It was neither. The O.P.M.S. Silver kratom product is a pure kratom leaf powder that consists solely of ground leaves from the tree, not an

extract kratom product.[1] There is no concentrated or synthetic levels of 7-hydroxymitragynine Mitragynine ("7-OH") in O.P.M.S. Silver. *Id.* Dr. Wharton is not familiar with how 7-OH is used in actual kratom products and testified 7-OH is "only produced in the human body." *Id.* at 43:1–47:13.

Dr. Wharton expressly admitted he is not a medical toxicologist and does not have any specific training or expertise in medical toxicology. *Id.* at 70:3-17, 100:3-23. He incorrectly testified that there was a level of 7-OH found in Ms. Moller's autopsy and that the presence of both mitragynine and 7-OH are "important to [his] opinion" as both mitragynine and 7-OH are "substances [] implicated in causing ventricular arrhythmias." *Id.* at 43:1–47:13. There was no mention of 7-OH in Ms. Moller's autopsy report by the medical examiner and 7-OH was *not present* on the primary toxicology report by NMS Labs (the only report Dr. Wharton recalled reviewing) and only found on the one-page UCSF toxicology report at a level of .8 ng/ml which is "below the lower limit of quantification of 3ng/mL and equal to the limit of detection." *Id.* at 43:1–47:13, 100:24-101:12.[2] Dr. Wharton admitted he does not know what a safe or lethal dose of mitragynine is. *Id.* at 71:22–72:25.

Dr. Wharton is not a pathologist. *Id.* at 67:18-70:5. Dr. Wharton has very limited experience in determining cause of death in his clinical work, only recalls signing death certificates as part of his training and has not had to certify a cause of death in over 10-15 years. *Id.*

Dr. Wharton stated in his Initial Report, it is his "understanding that Harmony Moller had no prodrome prior to her being found pulseless by her mother in the early morning hours" and that she "had no known signs or symptoms of chronic or acute disease or injury presaging her sudden

[1] *See* Exs. 3–4 (showing O.P.M.S. Silver kratom products at issue are not a liquid or an extract).
[2] *See* Ex. 8, St. Tammany Parish Coroner Office Records, at ST.TAMMANYCORONER_MOLLER00010-12, 14-19.

death." Ex. 1, Initial Report, at 2–3. But Dr. Wharton admitted he was not aware of Plaintiff statements that Harmony's suspected she had food poisoning, which he concedes can affect the heart through severe dehydration and electrolyte imbalance. Wharton Dep. at 122:24-124:3, 126:2-128:3. He further testified he does not "know Harmony Moller's physical state in the days and hours leading up to her death." *Id.*

Dr. Wharton did not review the police report, the fire department/EMS report, the over 150 crime scene photos or the body camera footage. Ex. 1, Initial Report, at 1; Ex. 2, Wharton Dep. at 74:10–77:10.

He admitted he was completely unaware of other items found in or around Harmony Moller's bedroom, including drug paraphernalia (foil, containers, grinder), vaping pens, alcohol, prescription bottles (most of which were not prescribed to her), dietary and herbal supplements, caffeine pills, marijuana products, energy weight loss supplements (*e.g.* Zantrex Black), and unidentified or unlabeled ground substances. *Id*. at 77:7–89:13. Dr. Wharton also admitted it would be important to find out what these other products were and that they would be "relevant." *Id.* at 93:15-17, 94:5-11, 144:4-9. For example, he "would want to know what she was vaping" because "if someone is taking a substance, you would want to know what the substance is because that's the only way you could determine what its potential for adverse effects are." *Id.* at 119:6-16.

Dr. Wharton admitted he would also "want to know which prescriptions this person was taking" and that he would be "alarmed" if she was taking a prescription not prescribed to her. *Id.* at 89:24-90:5. He testified he was aware that Harmony Moller had Phenergan (Promethazine) and

Benadryl in her system, but that he does not know whether the Phenergan was prescribed to her. *Id.* at 91:13-23. They were not.[3]

He stated that he "assume[d]" a "fairly extensive screen for anything—any substances that could possibly be in her system" but was "not aware" of what was tested or "screened for" in Mississippi where he (incorrectly) believed this case occurred. *Id.* at 92:13-25. Dr. Wharton was not familiar with the levels of toxicology panels that can be tested. *Id.* at 97:2-24.

Dr. Wharton stated in his Initial Report that there was no "family history of premature sudden cardiac death." Ex. 1, Initial Report, at 2. He was not aware that Harmony Moller's brother died prematurely in his thirties or anything about the cause of his death. Wharton Dep. at 101:20–103:22. He did not ask for any medical records and relied entirely on Plaintiff's counsel to determine what was relevant to his review. *Id.* at 104:17-5–106:18.

He admitted his opinion that kratom caused Ms. Moller's death is based on a "process of elimination" because he "can't think of anything else that would have done this." *Id*. at 216:2–11, 218:10–14.

Despite having no prior experience with kratom, toxicology, pharmacology, or pathology, Dr. Wharton summarily rejected the coroner's findings that Ms. Moller's primary cause of death was an "abnormally high left coronary ostia takeoff" that was only "exacerbated by mitragynine" as "incorrect." *Id.* at 16:21–5, 47:2–13, 70:3–19, 134:5–23. In his brief, three-page Initial Report, he provided no explanation why Ms. Moller's "abnormally high left coronary ostia takeoff" did not cause her death, only calling it a "benign finding which was not a cause of her sudden death." Ex. 1, Initial Report at 3. When asked why he thinks that, he responded that he does not "believe

[3] *See* Ex. 7, Excerpts from Deposition of Kathleen Moller, 227:15–229:2, Dec. 14, 2025.

it's [his] job to give [] a set of references" to support the conclusion that Ms. Moller's "coronary abnormality is benign." Ex. 2, Wharton Dep. at 136:12-137:21.

Thus, in his Initial Report, *without explanation*, Dr. Wharton concludes Harmony Moller's coronary anomaly had no role in her death and that it was "more likely than not, Harmony Moller died from a cardiorespiratory adverse event induced by consumption of kratom." Ex. 1, Initial Report, at 3.

In his deposition, Dr. Wharton admitted that all of his opinions in this matter are pure speculation.

> Q: In fact, in a lot of your opinions here today, you're speculating?
>
> A: We have no hard data on a drug - - or agent of the nature.
>
> Q: Your opinions Your opinions here today about the cause of death of Ms. Harmony Moller are based on speculation informed by literature and your experience as a cardiologist, correct?
>
> A: Yes.

Ex. 2, Wharton Dep. at 167:10–170:21. Dr. Wharton offered speculative testimony at multiple other occasions in his deposition. This includes "speculating" that the results of a recent FDA placebo controlled dosing study on kratom that demonstrated no serious adverse events from kratom consumption at much higher doses than in Ms. Moller's case, should be dismissed it involved participants who "speculat[ed]" were "otherwise perfectly healthy." *Id*. at 165:15–167:19. Dr. Wharton stated in his Rebuttal Report, without any explanation or analysis, "[i]t is certainly possible that the left ventricular hypertrophy noted in the autopsy was a manifestation of kratom exposure." Ex. 5 (January 2, 2026 Rebuttal Report of Dr. Ronald H. Wharton) ("Rebuttal Report"), at 2. Dr. Wharton admitted, this is possible "[o]nly if she been using it for a long time.

I have no idea how long she was using it." Ex. 2, Wharton Dep. at 227:10–228:7. He agreed that this was "just a speculation" as he does not know "one way or the other if it was caused by kratom" or "[i]t could just be from hypertension." *Id.*

His Initial Report claims his opinion is based on "multiple reports in the medical literature that associate the use of kratom products with the development of ventricular tachycardia and ventricular fibrillation (VT/VF), which cause the heart to stop beating." Ex. 1, Initial Report, at 2. His primary support for this conclusion relies on Krantz, M.J. et. al. Ventricular Arrhythmias Associated With Over-the-Counter and Recreational Opioids. *J Am Coll Cardiol 2023;81:2258–2268* ("Krantz Study"), (Ex. 6).[4] The Krantz Study is a retroactive analysis of publicly available adverse event reporting databases. *Id.* According to the Krantz Study, out of an estimated five million kratom users, there were 46 reports of VT/VF allegedly associated with kratom use. *Id.* Of these 46 reports, the level of alleged kratom ingestion is not known.[5] The Krantz study states "causal determinations cannot be made" and the ultimate conclusion is the opposite of Dr. Wharton's characterization. The Krantz study finds "***a lack of a convincing mechanistic link*** between mitragynine and ventricular arrhythmia." Wharton Dep. at 191:7-194:12 (emphasis added).

In his Rebuttal Report, Dr. Wharton revises his conclusion to stated "more likely than not, Harmony Moller died from a ventricular arrhythmia induced by consumption of Kratom,

[4] He also cites Miller et. al. Kratom Cardiotoxicity: Reversible Brugada Pattern and QTc Prolongation. *J Am Coll Cardiol Case Rep. 2025;30:103109* ("Miller Study"). The Miller Study is a single case report published in March 2025 involving a patient who was ingesting approximately 84 g to 100 g of kratom daily —a dose Dr. Wharton admitted was "[s]ignficiantly more" than the evidence suggests for Ms. Moller. Wharton Dep. at 206:9–16, 217:1–15.

[5] There are further limitations of these reports, including the possibility of duplicate reporting, a lack of verification and the inability to screen for the use of other drugs, in addition to kratom.

especially in combination with two other agents [promethazine and diphenhydramine] known to prolong the QT interval." Ex. 5, Rebuttal Report, at 2.

In his deposition, he testified "it's actually three other agents" as he "forgot the THC in [his] report" as THC "can raise the QT interval." Wharton Dep. at 130:10–132:16. He also stated that he does not recall if ondansetron (which Plaintiff admitted Harmony took before she died) causes QT prolongation, but "[i]f it does, we can add it." *Id.*

The literature that Dr. Wharton relies on to form his causation opinion does not support that kratom is an agent that causes QT prolongation. The Krantz Study "shows a complete lack of association of mitragynine to QT prolongation." *Id.* at 184:8–185:20.

The methodology in Dr. Wharton's Initial Report and Rebuttal Report rests on a foundation of ignorance, and his conclusions are pure speculation, and his opinion that kratom caused Ms. Moller's death should be excluded.

III. ARGUMENT

A. Standard for admissibility.

The district court has considerable discretion to admit or exclude expert testimony under Fed. R. Evid. 702. *See Gen. Elec. Co. v. Joiner*, 522 U.S. 136, 138–39 (1997); *Seatrax, Inc. v. Sonbeck Int'l, Inc*., 200 F.3d 358, 371 (5th Cir. 2000); *Tajonera v. Black Elk Energy Offshore Operations,* LLC, Civ. A. No. 13-0366, 2016 WL 3180776, at *8 (E.D. La. June 7, 2016) (Brown, J.) (citing authority). The burden rests with the party seeking to present the testimony to show that the requirements of Rule 702 are met. *Moore v. Ashland Chemical, Inc.*, 151 F.3d 269, 276 (5th Cir. 1998). Rule 702 provides that an expert witness "qualified . . . by knowledge, skill, experience, training, or education may testify in the form of an opinion" when the following requirements are met:

(a) The expert's scientific, technical, or other specialized knowledge will help the trier of fact to understand the evidence or to determine a fact in issue;

(b) The testimony is based on sufficient facts or data;

(c) The testimony is the product of reliable principles and methods; and

(d) The expert has reliably applied the principles and methods to the facts of the case.

Fed. R. Evid. 702. Rule 702 codifies the Supreme Court's opinion in *Daubert,* which charges district courts to act as "gatekeeper" when determining the admissibility of expert testimony. *Tajonera*, 2016 WL 3180776, at *8. The purpose of a *Daubert* motion is "to ensure that only reliable and relevant expert testimony is presented to the jury." *Rushing v. Kansas City S. Ry. Co*., 185 F.3d 496, 506 (5th Cir. 1999) (superseded by rule on other grounds) (citing *Daubert*, 509 U.S. at 590–93).

When an injury was allegedly caused by a substance, the Fifth Circuit applies a two-step framework to examine the admissibility of causation evidence. *Knight v. Kirby Inland Marine, Inc.*, 482 F.3d 347, 351 (5th Cir. 2007). First, the court must determine whether there is admissible evidence of general causation: that "a substance is capable of causing a particular injury." *Id.* Second, if the court "concludes that there is admissible general-causation evidence, [it] must determine whether there is admissible specific-causation evidence." *Id.* Critically, "[e]vidence concerning specific causation . . . is admissible only as a follow-up to admissible general-causation evidence." *Id.*

The Court's inquiry is whether the expert's reasoning is reliable at each step. *Knight*, 482 F.3d at 355. Opinions based on assumptions, speculation, or gaps between the data and the conclusion are inadmissible. *Id.*; *Viterbo v. Dow Chem. Co.*, 826 F.2d 420, 424 (5th Cir. 1987).

The proponent must prove admissibility by a preponderance, and "expert's assurances" do not substitute for objective reliability. *Moore*, 151 F.3d at 276.

B. Dr. Wharton is not qualified to offer his causation opinions in this case.

Dr. Wharton is a cardiologist. He is not trained in toxicology, pharmacology, forensic pathology, or kratom research. Yet, he proposes to tell a jury that kratom's active compounds caused a death. That opinion is outside his expertise, outside his training, and outside the limits Rule 702 imposes on expert testimony. Rule 702 does not allow a witness to leverage general medical credentials into authority over an entirely different scientific field.

An expert "may not go beyond the scope of his expertise in giving his opinion." *MGMTL, LLC v. Strategic Tech. Inst., Inc.*, 2022 WL 485279, at *6 (E.D. La. Feb. 17, 2022) (Vitter, CJ). Dr. Wharton does precisely that by offering opinions on the specific chemical properties of mitragynine and its effect on the human body—a subject he knew nothing about before this litigation, and still does not, as demonstrated in his deposition. His lack of expertise in the relevant fields is the first of many reasons his testimony should be excluded. *See Moore*, 151 F.3d at 276. In *Moore,* the Fifth Circuit affirmed the exclusion of an expert based on the district court's conclusion that the bases for their opinion was "individually and collectively inadequate" under *Daubert*. *Id.* at 277. The excluded expert's "training and experience . . . were obviously important to his diagnosis" but he "gave no reason why these items were helpful in reaching his conclusion on causation" and "admitted that he had never previously treated a patient who had been exposed to a similar Toluene solution" – the solution that was alleged to cause the harm. *Id.* at 277–78. The Court noted the excluded expert "was a highly qualified pulmonary specialist, but, as the Seventh Circuit observed in *Rosen v. Ciba-Geigy Corp.*, 78 F.3d 316 (7th Cir. 1996), "[u]nder the regime of *Daubert* a district judge asked to admit scientific evidence must determine whether the

evidence is genuinely scientific, as distinct from being unscientific speculation offered by a genuine scientist." *Id.* at 318 (internal citation omitted)."

Dr. Wharton is an accomplished cardiologist, whose expertise is in hypertrophic cardiomyopathy, a condition he opines Harmony Moller did not have and did not contribute to her death. Defendant is not challenging Dr. Wharton's expertise as a cardiologist or Dr. Wharton's qualifications to testify about the impact of Harmony Moller's "abnormally high left coronary ostia takeoff" or whether or not she had hypertrophic cardiomyopathy.

But Dr. Wharton has no knowledge, skill, experience of training in the core issues here: kratom, and its toxicology and pharmacology, and its ability to cause or contribute to death. Ex. 2, Wharton Dep. at 27:4–28:15, 70:3-17, 100:3-7. He expressly admitted he is not a medical toxicologist or pathologist. *Id.* at 27:4–28:15, 67:18-70:5.

During his deposition on February 23, 2026, the profound deficiencies in Dr. Wharton's qualifications and methodology became clear.

Dr. Wharton was not "familiar with kratom prior to receiving this case." Ex. 2, Wharton Dep. at 46:25–47:23. He has never done any research on kratom, never treated any patient for kratom exposure and never encountered kratom in his clinical practice. *Id*. His "opinions on kratom in this case are based solely on the articles and references that [he] cited in his [his] report." *Id.* at 59:7-11. To supplement his kratom knowledge he acquires information from the internet (e.g. on the "profit margin" on kratom) and is prepared to testify on those topics based solely on this research, when it is not his "job" "to give [] a set of references." *Id.* at 62:5–65:25; 137:6-21.

Dr. Wharton demonstrates a complete lack of knowledge on kratom products generally. He incorrectly referred to the O.P.M.S. Silver Products at issue as liquids and extracts – when they are pure kratom ground leaf material. He has no knowledge regarding how 7-OH is synthetically in

kratom products. He testified that the presence of 7-OH in Harmony Moller's autopsy is important to his opinion, but the coroner findings do not refer to any quantifiable level of 7-OH, nor is there any.

This is precisely the kind of expert overreach Daubert is designed to prevent. A cardiologist may testify about cardiac structure and disease. He may not step into the role of toxicologist simply because the case involves a heart. Dr. Wharton admitted he had never studied kratom, never treated a kratom exposure, never researched its pharmacology, and learned about it only after being retained for this case. Allowing him to testify anyway would present the jury with the imprimatur of medical authority untethered to actual expertise. Rule 702 requires more than credentials—it requires qualification in the subject at issue. Because Dr. Wharton lacks that qualification, his kratom causation opinions must be excluded.

C. Dr. Wharton failed to establish general causation before opining on specific causation.

Even assuming Dr. Wharton is qualified to address cardiac physiology, his general-causation opinions are independently inadmissible under Rule 702. Fifth Circuit precedent requires reliable scientific evidence identifying a harmful level of exposure before causation testimony may be presented to a jury. Dr. Wharton concedes that no such dose is known here. That failure is dispositive.

The Fifth Circuit requires admissible general-causation evidence before any specific-causation testimony may be admitted. *Johnson v. Arkema, Inc.*, 685 F.3d 453, 469 (5th Cir. 2012); *Knight*, 482 F.3d at 351. At a minimum, general causation requires scientific knowledge of a harmful level of exposure. *Johnson*, 685 F.3d at 469; *Allen*, 102 F.3d at 199. Dr. Wharton must therefore identify "the harmful level of exposure" to mitragynine that is capable of causing a fatal arrhythmia in the general population. *Kaoui v. BP Expl. & Prod., Inc.*, No. CV 17-3313, 2023 WL

330510, at *2 (E.D. La. Jan. 12, 2023) (Vitter, CJ), 2023 WL 330510, at *6. He has not, because he cannot. He admitted he does not know what constitutes either a safe or lethal dose. Ex. 2, Wharton Dep. at 71:22–72:25. Dr. Wharton acknowledged that his opinion is "based on reasonable speculation" and "his experience as a cardiologist" as "[w]e have no hard data on a drug or agent of this nature." *Id.* at 167:10–170:21.

Dr. Wharton claims that scientific literature suggests a "linkage between the main alkaloid in Mitragyna speciosa (kratom) use, ventricular arrhythmias (ventricular tachycardia, ventricular fibrillation) and cardiorespiratory adverse events" and "potentially fatal effects on the electrocardiogram attributed to the use of kratom." Ex. 1, Initial Report, 2–3. This opinion fails to establish general causation for several reasons. First, as noted above, the one article he cites for his opinion concludes "a lack of a convincing mechanistic link between mitragynine and ventricular arrhythmia" and state "causal determinations cannot be made." Ex. 2, Wharton Dep. at 191:7-194:12. Thus, the study does not identify the harmful level at which physical symptoms manifest though that was an object of the study. Thus, Dr. Wharton fails to cite evidence of general causation. *Kaoui*, 2023 WL 330510, at *6 ("To establish general causation, a causation expert must identify 'the harmful level of exposure to a chemical' at which physical symptoms manifest."). This Court has stressed, "nearly every chemical on Earth may be toxic or even fatal at a certain level of exposure," so "[c]ausation experts determine not only *whether* a chemical is capable of causing certain health effects, but *at what level* of exposure do those health affects appear." *Id*. "[F]ailure to properly identify the level of exposure to a particular chemical at which harmful effects occur necessarily renders a general causation opinion to be unreliable and, thus, inadmissible." *Id*.

Dr. Wharton's opinion on general causation is therefore unreliable and thus inadmissible because he fails to identify any evidence of a fatal dosage of kratom. *See also, Moore,* 151 F.3d at 277–79 (affirming district court's exclusion of expert where he had "no information on the level of exposure necessary for a person to sustain the injuries about which the MSDS warned. The MSDS made it clear that the effects of exposure to Toluene depended on the concentration and length of exposure."). Where assumptions substitute for facts—such as, for example, exposure, as here—causation devolves into "mere guesswork" and is inadmissible. *Seaman v. Seacor Marine L.L.C.*, 326 F. App'x 721, 727–28 (5th Cir. 2009) (affirming exclusion where assumptions about exposure levels rendered causation opinion "mere guesswork" and failed to establish either general or specific causation).

Even assuming Dr. Wharton is qualified in his own field, his general-causation opinions must be excluded because they fail the Fifth Circuit's most basic reliability requirement: identifying a scientifically established harmful level of exposure.

D. Dr. Wharton's specific causation opinion is unreliable.

Even if Dr. Wharton could establish general causation—which he cannot—his specific causation opinion independently fails under Rule 702 because it rests on a fundamentally unreliable "process of elimination." He admitted that he concluded kratom caused Harmony Moller's death because he "can't think of anything else that would have done this." Ex. 2, Wharton Dep. at 218:10–14. That is not science. It is guesswork.

1. The Fifth Circuit has rejected this exact methodology.

An expert may eliminate alternative causes only after establishing that the substance at issue is capable of causing the injury and only if the elimination itself is performed in a scientifically reliable manner. Where either step fails, the opinion is inadmissible.

Dr. Wharton characterizes his method as a "differential diagnosis," which he agrees requires systematically eliminating "any other possible cause." Ex. 2, Wharton Dep. at 248:14–249:24. But the Fifth Circuit has made clear that process-of-elimination reasoning is not admissible unless it is anchored in reliable scientific groundwork. In *Black v. Food Lion, Inc.*, the Fifth Circuit excluded expert testimony where the expert concluded that because she had ruled out other possible causes, the remaining event "had to be" the cause. 171 F.3d 308, 313 (5th Cir. 1999). *Id.* The Court explained, "This analysis amounts to saying that because [the expert] thought she had eliminated other possible causes of [the condition], even though she does not know the real 'cause,' it had to be the [injury-causing event]. This is not an exercise in scientific logic but in the fallacy of post-hoc propter-hoc reasoning." The Fifth Circuit further held that such a "general methodology cannot vindicate a conclusion for which there is no underlying medical support." *Id.* at 314.

That is precisely what happened here. Dr. Wharton does not identify a scientifically established harmful dose of mitragynine. He does not establish a validated mechanism of death tied to a known exposure level. Instead, he assumes kratom must be responsible because he does not see a better explanation. That is the same reasoning rejected in *Black*.

A differential diagnosis is not a free pass. It is admissible only when the expert reliably establishes general causation and then methodically rules out alternatives based on complete and accurate data. It is not admissible when it functions as a fallback after scientific proof fails.

2. Dr. Wharton did not reliably rule out obvious alternative causes.

Even if process-of-elimination reasoning were theoretically permissible here, Dr. Wharton did not perform it reliably. He acknowledged that both promethazine and diphenhydramine were present in Harmony Moller's system—drugs known to prolong the QT interval—yet he offers no

scientifically grounded basis for concluding kratom was the decisive agent. He admitted he had not reviewed critical materials, including crime scene photographs showing numerous substances present at the scene. Those substances included vaping devices, alcohol, prescription medications, supplements, caffeine products, marijuana products, and unidentified materials. He testified that he "would want to know what the substance is because that's the only way you could determine what its potential for adverse effects are," yet he did not know what many of those substances were. That alone defeats any claim of systematic elimination.

He further admitted he "forgot the THC" in his report. When asked about ondansetron—another drug potentially associated with QT prolongation—he stated that "[i]f it does, we can add it." Wharton Dep. at 130:10–132:16. That is not a defined methodology. That is improvisation.

His approach effectively operates as a one-way ratchet: any substance known to prolong QT can be added to support causation, but none is rigorously evaluated as an independent cause. Meanwhile, kratom is treated as sufficient without a scientifically established dose-response foundation.

3. Dr. Wharton failed to investigate critical medical and physiologic factors.

Dr. Wharton also relied on assumptions that collapse under scrutiny. He asserted that there was no relevant family history of premature sudden cardiac death, yet he was unaware that Ms. Moller's brother died unexpectedly in his 30s and does not know the cause. For a cardiologist opining on arrhythmia, failure to investigate family cardiac history is not a minor omission—it is a fundamental methodological flaw.

He also assumed Harmony Moller had "no recent medical illnesses and no chronic medical conditions," yet admitted he did not know her physical condition in the days and hours before her death. Records reflect reports of vomiting and possible food poisoning—conditions known to

cause electrolyte imbalance and trigger ventricular arrhythmias. He acknowledged that such electrolyte disturbances can precipitate arrhythmias. Yet he did not analyze or rule out this pathway.

That is not systematic exclusion. It is selective omission.

4. Dr. Wharton's approach is guesswork disguised as expertise.

Rule 702 requires intellectual rigor. *Kumho Tire Co. v. Carmichael*, 526 U.S. 137, 152 (1999). It requires the court to ensure reliability at every step. *Knight*, 482 F.3d at 355. Dr. Wharton did not establish a harmful kratom exposure threshold, did not reliably quantify competing substances, did not investigate known arrhythmia triggers, did not review critical medical history, was not aware of Harmony's physical condition prior to death, and relied on assumptions about toxicology screening.

When elimination rests on incomplete information and untested assumptions, it is not differential diagnosis—it is "mere guesswork." *Seaman*, 326 F. App'x at 727–28. The Fifth Circuit has already warned against exactly this maneuver: using temporal sequence and absence of a preferred alternative as proof of causation. *Black*, 171 F.3d at 313–15. That is what Dr. Wharton has done here. His specific causation opinion should be excluded.

5. Dr. Wharton's approach fails Daubert.

Even apart from the Fifth Circuit's squarely on-point holding in Black, Dr. Wharton's methodology collapses under the core Daubert reliability factors. What remains is not scientific reasoning but subjective belief dressed as expert testimony. *Moore*, 151 F.3d at 277–79 (affirming exclusion where "[n]one of Daubert's factors to assess whether the opinion was based on sound scientific principles was met"). In *Moore*, the expert could cite "no scientific support" for the proposition that exposure to irritants at unknown levels caused the alleged harm. The Fifth Circuit

held that such speculation "lacking any scientific validity" must be excluded. *Id.* That principle applies with equal force here.

First, Dr. Wharton's core theory—that a post-mortem mitragynine concentration of 150 ng/mL alone caused Ms. Moller's death while superseding her documented coronary anomaly and the presence of other substances—is a litigation-generated hypothesis. It has never been tested. It cannot be tested. There is no established exposure threshold, no validated dose-response curve, and no empirical method by which his conclusion could be verified or falsified.

Second, his dismissal of Ms. Moller's "abnormally high left coronary ostia takeoff" as a "benign finding," (Ex. 1, Initial Report, at 3), despite the coroner's contrary conclusion, is not grounded in peer-reviewed authority. He has identified no literature validating that conclusion under these circumstances. When pressed in deposition, he conceded that it was not his "job" to provide supporting references. That is not expert methodology. It is assertion. Daubert requires more than credentials and confidence.

Third, there is no identifiable protocol for testing Dr. Wharton's case-specific, multi-variable theory. Without a defined methodology, there is no measurable rate of error. There are no governing standards. There is no control against confirmation bias. Those are classic markers of courtroom-driven speculation, not reliable science.

Fourth, even accepting differential diagnosis as a generally accepted clinical tool, Dr. Wharton did not apply it reliably. A proper differential diagnosis requires systematic consideration and exclusion of all plausible causes. He failed to do that. He did not fully evaluate the independent arrhythmogenic potential of other QT-prolonging drugs present. He admitted he "forgot" certain substances and was willing to "add" others after the fact. He did not meaningfully assess the decedent's vomiting and possible dehydration—well-known triggers of electrolyte

imbalance and arrhythmia. He did not investigate the significance of a family history of sudden premature death. His process was not structured elimination; it was selective omission.

Finally, even if differential diagnosis is generally accepted in medicine, this version of it is not. The relevant medical community would not accept a causation opinion that: (1) lacks a known harmful dose, (2) ignores competing pharmacologic effects, (3) fails to evaluate documented physiologic stressors, and (4) dismisses structural cardiac findings without supporting literature. General acceptance does not attach to an expert's flawed application of a method. It attaches only to a reliable execution of it.

Dr. Wharton has offered a conclusion in search of support. Daubert does not permit that.

6. Because Dr. Wharton's general causation opinion is inadmissible his specific causation opinion must also be excluded.

Dr. Wharton's specific causation opinion is inadmissible as a matter of law because "[e]vidence concerning specific causation in toxic tort cases is admissible only as a follow-up to admissible general-causation evidence." *Knight*, 482 F.3d at 351.

As established in Section III.C, Dr. Wharton's general causation opinion is inadmissible because he failed to establish the requisite dose-response relationship. He admitted he does not know what a safe or lethal dose of mitragynine is. Ex. 2, Wharton Dep. at 71:22–72:25. Because he cannot clear the general causation hurdle, his specific causation opinion must be excluded.

IV. CONCLUSION

For these reasons, Defendant respectfully request that the Court exclude Dr. Ronald H. Wharton's causation opinions in their entirety,[6] including any opinions that: (1) kratom caused

[6] While Defendant agrees Dr. Wharton has the expertise to opine on whether or not Harmony Moller had hypertrophic cardiomyopathy, considering Dr. Wharton's failure to satisfy the Daubert factors, as described herein, Defendant requests Dr. Wharton's opinion be excluded in its entirety.

Harmony Moller's death; (2) the reported mitragynine concentration was "sufficient" to cause adverse events or death; and (3) other potential causes or contributors were reliably ruled out.

Dated this 2nd day of March, 2026.

Respectfully Submitted

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CERTIFICATE OF SERVICE

This is to certify that on the 2nd day of March, 2026, a copy of the foregoing document was filed with the Clerk of Court and served on all counsel of record via the Court's ECF System.

/s/ Gwendolyn C. Payton
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